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Improving agricultural land use detection through YOLO–SAM fusion framework

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ABSTRACT

Agricultural land management increasingly depends on accurate object detection and segmentation from aerial imagery. Traditional approaches relying on bounding boxes often lack the precision required for detailed agricultural monitoring, whereas manual segmentation remains time-consuming and costly. This study develops a YOLO-SAM fusion framework integrating the You Only Look Once (YOLO) detector with the Segment Anything Model (SAM) to enhance the accuracy and efficiency of agricultural field detection and delineation from high-resolution drone imagery. The proposed model uses YOLO's bounding-box centroids as input prompts for SAM, linking semantic recognition with geometric segmentation to generate precise polygon boundaries without manual annotation. The framework also enables direct integration into QGIS for visualization and spatial analysis, thereby supporting practical GIS-based applications. The YOLO-SAM fusion model achieved optimal performance with a confidence score of 0.5 and 80% tile overlap during image tilling, resulting in a detection area ratio of 96.5% and an error area ratio of 4.0%. These results demonstrate the proposed approach effectively manages complex agricultural environments while reducing data preparation costs and providing a scalable, GISoperational solution for automated agricultural field mapping and resource management.

1. Introduction

Evidence-based regional planning for sustainable development relies heavily on precise geographic information systems (GIS) and effective resource management [13,33,35]. Within this framework, accurately identifying and extracting key geographical features such as roads, facilities, and agricultural lands are crucial to support decision-making processes in regional planning, agricultural management, and environmental monitoring [24,29]. Automated object detection based on large-scale imagery enhances monitoring and management possibilities, optimizing resource allocation and minimizing environmental impacts [6,42].

You Only Look Once (YOLO) is a widely used deep learning model in object detection, noted for its high accuracy and fast processing speeds [30]. Particularly in agriculture, recent research has used deep learning-based object detection, especially YOLO algorithm, to recognize flowers and fruits for growth modeling and diagnosing the types and reproductive status of pests for digital agricultural technologies [4,10,37,43]. Ruan et al. [31] reported using the YOLO algorithm to detect crop rows in rice seedlings, aimed at unmanned agricultural machinery applications. However, YOLO primarily employs bounding boxes (bbox) for object identification, which presents limitations for precise boundary segmentation required in crop monitoring. Developing advanced YOLO methodologies could overcome these limitations and effectively manage environments with high variability, such as agricultural lands and urbanizing areas. Despite recent advancements in the YOLO algorithm include segmentation techniques, they require polygon-shaped training data, which is more costly and time-consuming to prepare than bbox data [3,25]. Although YOLO-based segmentation models (e.g., YOLOv8-seg) can use mask annotations that may be generated semi-automatically, preparing accurate segmentation data for agricultural imagery remains highly time-consuming. Therefore, an automated segmentation approach capable of refining bounding-box detections into accurate object boundaries is required.

While recent studies have continuously enhanced YOLO's architecture for object detection across diverse domains, their focus has

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primarily been on structural improvements rather than modular integration. For instance, Jiang et al. [18] and Li et al. [22] optimized YOLOv8 by embedding attention and convolutional refinements (C2f-SCConv, SEv2) to enhance accuracy and efficiency, while Wang et al. [39] adapted YOLO for underwater imaging through background suppression modules. In agricultural contexts, models such as CM-Net for winter wheat [38], YOLO-PDGT for pomegranate detection [12], and a two-pathway segmentation network for rice row mapping [8] have demonstrated domain-specific advances. Yet, they remain dependent on polygon-based annotations and task-specific datasets. Other approaches, including Qureshi et al. [28] and Sun et al. [34], explored multimodal fusion using aerial or multispectral data, but did not address the challenge of generating semantically meaningful and geometrically precise boundaries at the parcel scale.

The Segment Anything Model (SAM), developed by Meta AI Research in 2023, is an emerging application based on deep learning in transformer-based image encoder and recognition tasks to extract image features and generate segmentations [5,36]. SAM is an object segmentation model that automatically segments various objects without specific prior training, accepting inputs in various forms (points, bboxes, text prompts) and offers high accuracy and precision. It is applied to observation-driven studies such as spatially small as well as irregular boundary conditions in any domain such as a medical environment and glaciology [27,32,41]. The performance of SAM was revealed by comparing evaluation methods among seven segmentation algorithms that accurately identify interesting objects of various sizes with rich semantic content and high consistency with digitized reference polygons [14]. However, while SAM provides exceptional segmentation capabilities, its inability to assign semantic meaning to segmented objects limits its standalone effectiveness in agricultural monitoring applications [23].

The increasing complexity of agricultural monitoring demands a solution that can bridge the gap between efficient object detection and precise boundary segmentation. An integration of YOLO's semantic understanding with advanced segmentation capabilities could potentially address these challenges while maintaining computational efficiency. For this integration, known bbox attribute data are used to apply YOLO for object recognition, and the recognized bbox centroids are used as input points for SAM, which helps assign meaning to the objects. The refined segmentation data are then utilized in the YOLO-Seg model to detect objects in large-area images.

Previous studies have made important contributions in boundary delineation, abandonment detection, and monitoring land-use change, but attempts to systematically integrate deep learning-based object detection with precise segmentation for parcel-level agricultural mapping remain limited. Chen et al. [7] and Cheng et al. [9] proposed rule-based methods to delineate farmland protection zones, relying primarily on administrative criteria and spatial connectivity rather than automated image-based detection. Hong et al. [15] utilized remote sensing to identify patterns and drivers of cropland abandonment in mountainous regions, yet their focus was on broader land-use transitions rather than fine-scale boundary extraction. Duan et al. [11] combined high-resolution multispectral imagery with hybrid modeling to predict land-use change, however, their analysis emphasized large-scale dynamics, offering limited applicability for precise field-level segmentation. Hosseini et al. [16] highlighted the potential of AI-driven land administration systems but concentrated on institutional and managerial applications rather than object-level detection. Similarly, Iilonga and Ajayi [17] applied deep learning to model land-use dynamics but did not address the integration of object recognition with boundary segmentation.

Many studies demonstrate the value of agricultural monitoring but fall short of providing methodologies that simultaneously deliver accurate object detection and precise boundary delineation. Wu and Dong [40] have combined detection and segmentation networks to address this challenge, such as integration of YOLOv5 and U-NET for crop type classification and field identification. However, these approaches remain sequential rather than modular, resulting in inconsistencies between semantic detection and geometric segmentation. YOLO offers efficiency and semantic recognition capabilities but often produces coarse boundary estimates, while SAM excels at generating accurate segmentations but lacks the ability to assign semantic meaning. A fusion approach that integrates YOLO's object recognition with SAM's segmentation strengths can address this gap, enabling both semantically meaningful and geometrically precise delineation of agricultural parcels.

The main goals of this study are to develop a modular YOLO-SAM fusion framework that sequentially links YOLO's semantic detection with SAM's geometric segmentation through centroid prompt interaction. This structure eliminates the need for costly polygonal training data by utilizing existing bounding-box annotations and automatically refining object boundaries. Moreover, the integration into a QGIS-compatible framework facilitates direct visualization and spatial analysis for agricultural field detection. The proposed workflow thus provides a cost-efficient and scalable approach to achieve high-precision parcel delineation while ensuring operational interoperability within GIS environments.

2. Materials and methods

2.1. YOLO algorithm

The YOLO algorithm is a methodology used for object detection targeting images or video streams. It is particularly known for its rapid processing speeds and high accuracy. The fundamental model of the YOLO algorithm utilizes a single neural network structure that divides the image into multiple regions, predicting bboxes and class probabilities for each area. Since the initial open-source release of version 1 in 2015, the YOLO algorithm has continuously improved [2], with updates extending to version 9 as of 2024. We adopted YOLOv8m for this study, referring to its reported performances on the common objects in context (COCO) benchmark (mAP 50.2, inference speed = 234.7 ms on CPU ONNX, and 1.83 ms on A100 TensorRT), which is a large-scale object detection, segmentation, and captioning dataset [19]. The benchmark performance was used to select the model variant, not to represent the results of our own training. The model segments the image into a grid with each grid cell predicting the probability of containing the center of an object, the coordinates of the bbox surrounding the object, and the class of the object.

2.2. Segment Anything Model (SAM)

SAM is an innovative deep-learning model used for object segmentation. It was initially released as open-source code. SAM is designed to address the challenges of object segmentation and has proven to be superior to other models [21]. It was trained on the SA-1B dataset, which includes over one billion masks derived from eleven million images, and boasts the capability of zero-shot generalization, enabling it to adapt to new segmentation tasks without additional training.

SAM generates masks automatically through an embedded mask decoder algorithm when prompted with inputs such as points or lines (Fig. 1). When prompts are ambiguous, the model can produce multiple valid masks to represent possible object boundaries. In this study, since only centroid points derived from YOLO bboxes were used as prompts, three candidate masks were generated by default. Among these, the mask with the largest IoU value against the detected bbox was automatically selected as the final segmentation output. This approach minimizes ambiguity while preserving the geometric accuracy of field boundaries.

SAM also generates masks without the need for input annotations. As a result, while the current SAM model can accurately detect objects, direct input is required to define what the objects are. To overcome this limitation, leveraging pre-existing annotated data could facilitate more

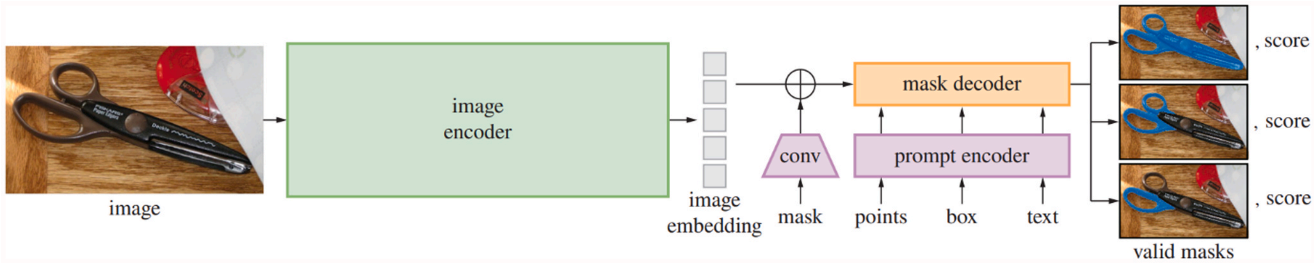

Fig. 1. Segment Anything Model (SAM) overview.
Source: [21]

accurate object identification. Therefore, this study aims to integrate annotation information from objects detected by the YOLO algorithm with SAM. This process will enhance the utility of the model for specific objectives.

2.3. Study process

We developed a process combining the deep learning-based YOLO algorithm with SAM to semi-automatically extract segmentation training data, augment this data, and ultimately extract segmentation objects (Fig. 2). The process was designed as a modular workflow linking YOLO's semantic detection with SAM's geometric segmentation to improve precision and automation efficiency. The research commenced with initial YOLO training and prediction using existing training data to identify the fundamental structure and positioning of objects within the images as bboxes. The centroids of these bboxes were then used as input prompts for SAM and extracting segmentation information. Given the potential for discrepancies due to ambiguous input prompts, the SAM-derived results were visually inspected in a semi-automated quality assurance (QA) step to ensure segmentation accuracy. This inspection process involved rapid binary decisions (accept or reject) without manual polygon editing, thereby maintaining the overall automation of the pipeline.

The final phase involved using the SAM-constructed segmentation data to retrain the YOLO algorithm for segmentation learning and object detection, thus forming a self-improving loop between detection and segmentation. After model inference, the YOLO–SAM fusion results were converted into georeferenced shapefile (SHP) format, enabling integration with GIS tools such as QGIS for spatial visualization, validation, and comparison with reference data. This comprehensive workflow provided a reproducible and practical framework for automated object detection, segmentation, and geospatial applications.

The YOLO–SAM fusion model was trained and tested on drone imagery using an NVIDIA RTX 5090 GPU (32 GB memory) with 32 GB RAM under Windows 11. The YOLOv8m detector was trained for 200 epochs with a learning rate of 0.001 using the Adam optimizer, and the default YOLO loss function combining classification, bounding box, and segmentation components. Data augmentation included random vertical and horizontal flips and rotations of 15° and 90°. The SAM model (vit-h) utilized pre-trained weights without additional fine-tuning. All experiments were implemented in Python 3.10, based on the PyTorch 2.0.1 and CUDA 12.1 frameworks, and integrated with QGIS 3.34 through the PyQGIS API for geospatial processing.

2.4. Image acquisition

We collected drone images from rural areas to identify and extract paddy fields, thereby constructing the training data. The captured drone images featured a resolution of 8 cm and covered a total area of 15 km^2 (Fig. 3). Due to the high resolution of the entire dataset, analyzing all the drone images was challenging; therefore, the images were split into manageable sections of 1024 × 1024 pixels for analysis. During the preparation of the training dataset, a 10 % tile overlap was applied to ensure that edge features were not lost when generating grid-based training samples. This overlap setting was used only for constructing the labeled training data. In contrast, the inference and evaluation experiments later tested multiple overlap ratios from 0 % to 90 % to determine the optimal configuration for accurate field detection and error minimization.

Furthermore, the study aimed to compare the results predicted by

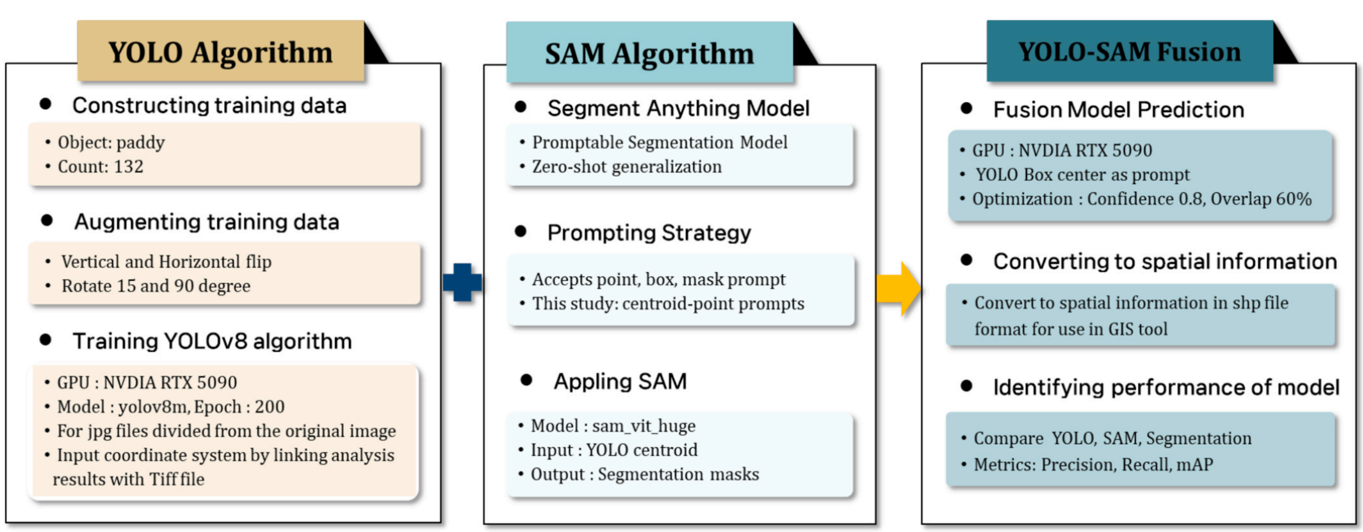

Fig. 2. Workflow for the YOLO-SAM Pipeline for Automatic Paddy Detection. The pipeline includes YOLO-based detection, SAM-based segmentation refinement, and spatial conversion to QGIS-compatible shapefiles for georeferenced analysis.

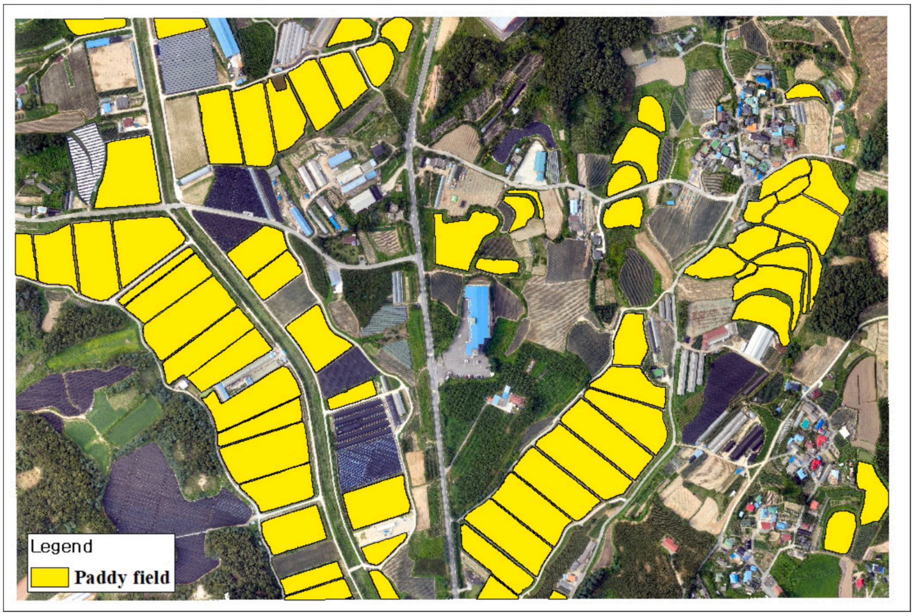

Fig. 3. Paddy field boundary based on the digital map and drone images in the study area.

each model. To facilitate this comparison, a preliminary visual survey was conducted to spatially document the paddy fields in the study area, which served as reference data for comparing the outcomes of each model. The surveyed paddy fields comprised 71 plots with a total area of 0.2336 km^2. All samples were collected from a single rice-growing district, representing typical field geometries and surface conditions in the region. This approach provided a structured method for data analysis and established a robust framework for evaluating the effectiveness of different models in predicting and mapping agricultural land.

2.5. Evaluation metrics

Object detection performance is evaluated based on precision, recall, and average precision (AP) metrics. Precision refers to the ratio of correctly identified positives relative to the total predicted positives (Eq. (1)).

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

The criteria for determining true positives and false positives are based on the intersection over union (IoU). The IoU is calculated as the ratio of the area of overlap between the ground truth bbox and the predicted bbox to the area of the union of these two boxes. Assessing the IoU value determines whether the predicted objects match the ground truth. In this study, the default YOLOv8 configuration was used without manual modification. Although lower IOU thresholds (e.g., 0.25) are sometimes used internally during training to facilitate loss computation or warm-up stages, the evaluation of detection performance employed the standard IoU = 0.5 threshold, consistent with YOLOv8′s default setting.

Furthermore, recall is defined as the ratio of actual true positives that the model correctly predicts as true. It is calculated using the following equation (Eq. (2)).

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

Lastly, the AP is calculated as the area under the precision-recall curve (PRC). This metric measures how well precision is maintained as recall increases, making it a key indicator for evaluating the performance of a model. In scenarios where multiple objects are detected simultaneously, the mean average precision (mAP) is computed to assess the effectiveness of the model (Eq. (3)).

$$mAP = \frac{1}{N}\sum\nolimits_{i=1}^{N} AP_i \tag{3}$$

3. Results

3.1. Accuracy of paddy field extraction through YOLO bboxes

We employed the YOLO algorithm to extract paddy fields using drone-captured imagery. Images were segmented into 1024 × 1024 pixel sizes and annotated with bboxes around the paddy areas. Data augmentation techniques, such as vertical and horizontal flips and rotations of 90 and 15 degrees, were applied to enhance model accuracy, resulting in 914 training samples. These samples were then divided into training and validation datasets in an 80:20 ratio. We used the YOLOv8m model and trained over 200 epochs on a GPU.

The precision of the model improved significantly with repeated training, reaching a maximum of 0.969, while the recall increased to 0.961. Furthermore, the analysis of the AP for predicted results revealed that when the IoU was above 50 %, peaked at 0.969, and for an IoU ranging from 50 to 95 %, the AP rose to 0.821 (Fig. 4). Detailed epoch wise segmentation metrics are provided in Table S1 of the Supplementary Information (SI).

Using the weight files trained through the YOLO algorithm, object detection was performed on the split images of the study area (Fig. 5). The parameters were set with a confidence threshold of 0.25, an overlap ratio of 10 %, and an IoU threshold of 0.5. Most bounding boxes (about 287 bboxes) were predicted, among which 21 were identified as false detections that did not correspond to actual agricultural fields. These errors typically occurred in areas where narrow drainage ditches in crop fields did not appear in the images, leading to the misclassification as paddy fields (Fig. 6). This pattern of false positives suggests a potential domain gap caused by limited diversity in the training data. Because the reference samples were collected from a single rice-growing district under similar environmental and lighting conditions, the model occasionally misinterpreted visually similar textures, such as grasslands or narrow furrowed areas, as paddy fields.

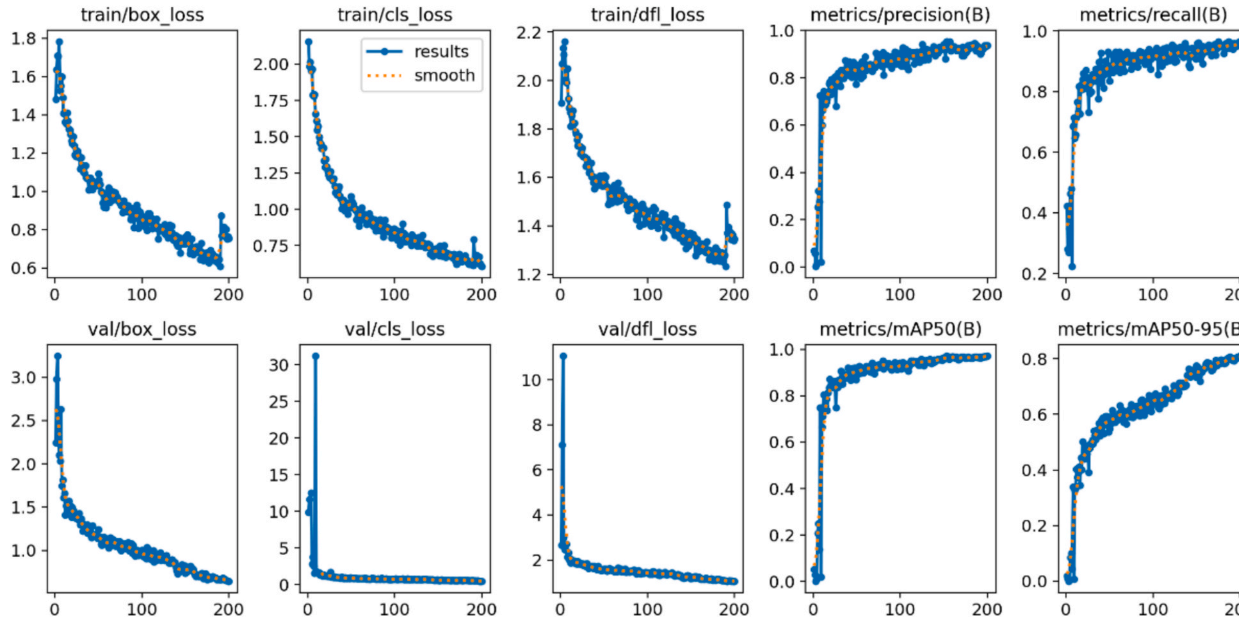

Fig. 4. YOLO training performance with bbox training data. The first three plots show the decreasing trends of training losses (box, classification, and distribution focal loss), indicating stable convergence over 200 epochs. The subsequent plots display validation losses, which follow a similar downward trend, confirming strong generalization performance. The rightmost panels present the evolution of precision, recall, and mean Average Precision (mAP) for bounding-box evaluation, where all metrics gradually improve and stabilize as training progresses.

Fig. 5. BBox results for Predict/Detect on paddy field using weights trained based on YOLO. The red polyline is the boundary of the actual paddy field; green is the region of the YOLO predicted results that contains the actual paddy field; and blue is the region of the YOLO predicted results that does not contain the actual paddy field.

3.2. Accuracy of paddy field extraction through YOLO-SAM fusion model

SAM was employed to extract the segmentation boundaries using the object information detected by YOLO. The centroids of the YOLO bboxes were used as input points for SAM. The parameters applied were identical to those used in the YOLO bounding-box prediction: confidence threshold = 0.5, overlap ratio = 10 %, and IoU threshold = 0.5. A total of 5,334 segments were generated, and 0.210 km^2 of the actual agricultural area (0.234 km^2) was correctly identified, yielding an accuracy of 89.7 %. The false-positive area was 0.034 km^2, accounting for 9.3 % of the total detected area. In the initial experiment, the YOLO–SAM fusion model achieved an accuracy of 89.7 % (Fig. 7).

Upon further analysis of the individual plots, it was found that 51 plots had an error margin within 10 % between the actual paddy areas and those extracted through YOLO-SAM fusion model. In comparison, 8 plots showed a discrepancy of more than 30 %. This analysis highlights the effectiveness of integrating YOLO with SAM for more precise agricultural field segmentation while noting areas where accuracy can improve.

Fig. 6. BBoxes predicted/detected via YOLO were detected in non-agricultural areas. The left image is a field area, but it was recognized as a rice field due to the narrow width of the furrows for drainage. The right image is a grassland area, but it was recognized as a rice field due to its similar shape and color features.

Fig. 7. Paddy Field Segmentation Information Extracted through YOLO-SAM Fusion Model (Confidence score = 0.5, Overlap = 10 %). The green polygons represent the boundaries of the fields extracted by YOLO–SAM, while the red lines indicate the ground-truth field boundaries derived from reference data. The strong spatial correspondence between the model-derived and actual boundaries demonstrates the model's capability to accurately delineate agricultural parcels. Minor discrepancies observed along irregular edges and narrow drainage ditches primarily result from geometric complexity and local texture similarity.

3.3. Optimal performance of YOLO-SAM fusion model

The performance of YOLO-SAM can be further improved by optimizing the object detection parameters of the YOLO model. In YOLO, the confidence score determines the threshold for recognizing objects, whereas an inappropriate setting can lead to two opposite outcomes: a low confidence score may include undesired non-agricultural areas, while a high confidence score may cause true agricultural fields to be missed. Additionally, the overlap ratio between adjacent tiles helps maintain spatial continuity in image analysis. Increasing overlap can reduce boundary omission errors but simultaneously raises the likelihood of duplicate detections and computational redundancy. To identify the optimal parameter configuration, this study systematically varied the confidence score (0.1–0.9 in increments of 0.1) and overlap ratio (0–90 % in increments of 10 %), resulting in a total of 90 parameter combinations, which were compared against the ground-truth reference data.

The analysis revealed that the two parameters influenced detection accuracy in distinct ways. As shown in Fig. 8, the confidence score directly determines the detection threshold for paddy fields, influencing both precision and sensitivity, whereas the overlap ratio controls the degree of redundancy by adjusting the number of overlapping image tiles. Although higher overlap helps capture missing boundary areas, it can also accumulate false detections across repeated tiles. Fig. 8a shows that as the confidence score increased, the total detected area decreased. At low confidence levels (0.1–0.3), the median detected area ranged between 0.3–0.4 km^2, substantially exceeding the actual cultivated area (0.2336 km^2) due to over-detection, while variation across overlap settings remained high. At confidence = 0.8, the median detected area closely matched the ground-truth area with reduced variability, indicating stable detection performance. However, at confidence = 0.9, the detected area dropped sharply below 0.1 km^2, reflecting severe under-detection.

The TP area analysis (Fig. 8b) showed stable detection between

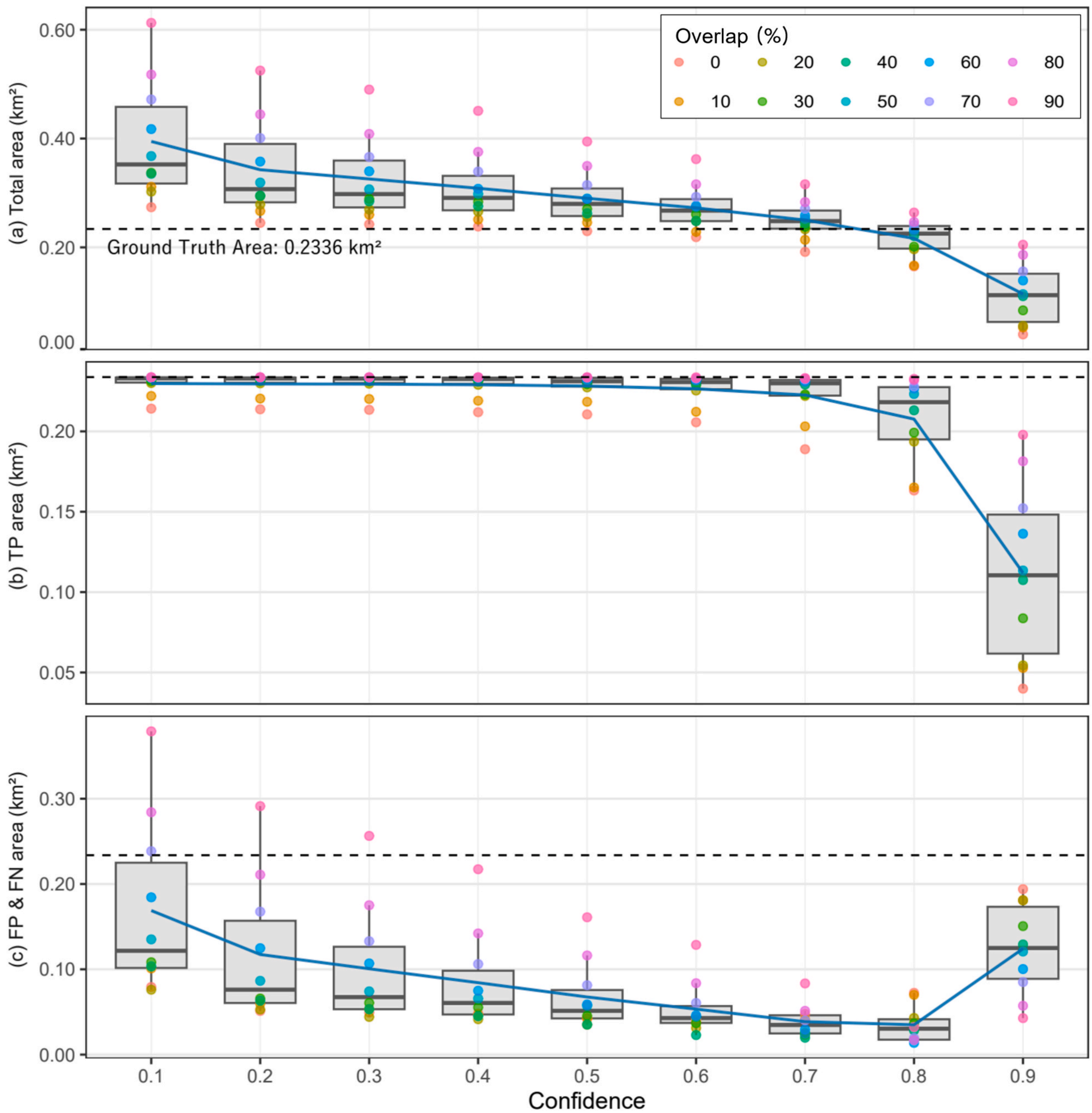

Fig. 8. Relationship between confidence score, overlap ratio, and detection performance in the YOLO–SAM model. (a) Total detected area shows a decreasing trend as the confidence score increases, with low confidence levels (0.1–0.3) producing substantial over-detection relative to the true paddy field area (0.2336 km^2), while higher confidence levels (>0.8) lead to under-detection. (b) TP area remains stable between confidence values of 0.1 and 0.7, maintaining close agreement with the reference area, whereas a sharp decline occurs beyond 0.8, indicating missed detections. (c) FP and FN areas demonstrate an inverse relationship, FP dominates at low confidence values, but FN increases significantly when confidence exceeds 0.8. In addition to accuracy, the computational cost associated with increasing the overlap ratio was also analyzed (See Table S2 in the SI for the detailed information). As the overlap increased from 10 % to 90 %, the processing time rose exponentially from approximately 6.5 s to 1,488 s per image and the number of tiles processed expanded from 234 to 15,656. This growth directly impacts both memory usage and GPU load, since a higher overlap requires reprocessing of redundant image regions across multiple tiles. Although larger overlap values slightly improve boundary accuracy by reducing omission errors, the substantial increase in time and memory demands makes them impractical for large-scale or near real-time applications. Therefore, considering both accuracy and computational efficiency, an overlap between 60 % and 70 % provides an optimal balance for operational use.

confidence = 0.1–0.7, where TP areas ranged from 0.22–0.23 km^2, closely matching the true field area. In this range, the narrow boxplot distribution indicates consistent detection regardless of overlap variation. Beyond confidence = 0.8, the TP area began to decline, and at confidence = 0.9, it dropped below 0.1 km^2, missing most agricultural regions. The FP and FN area analysis (Fig. 8c) revealed a clear transition between error types. At low confidence scores (0.1–0.3), FP dominated with median values exceeding 0.1 km^2 and maxima surpassing 0.3 km^2 as overlap increased. Conversely, for confidence $\geq$ 0.7, FP areas decreased sharply (<0.05 km^2), but FN areas rose notably at confidence = 0.9, indicating under-detection. Notably, the range confidence = 0.7–0.8 produced both low FP and FN areas, minimizing overall error. These results highlight that optimal detection accuracy and minimal error can be achieved through balanced parameter tuning, particularly around confidence = 0.7–0.8 and overlap = 70–80 %, where the YOLO–SAM model consistently detects agricultural parcels with high precision and reduced false detections.

Through comprehensive parameter optimization, the combination of confidence score = 0.8 and overlap = 60 % was identified as the optimal configuration. Under these conditions, the detected area reached 0.2334 km^2, corresponding to 99.85 % of the actual cultivated area, with a Recall of 96.98 %, Precision of 97.13 %, and an F1-Score of 97.05 %, demonstrating exceptionally high performance. As shown in Fig. 8, the boxplots for confidence score 0.8 closely align with the target values while exhibiting minimal variation across overlap levels, indicating stable and consistent detection accuracy.

Among the top five parameter combinations ranked by F1-Score, four occurred at confidence score 0.8, and the overlap 60 % configuration achieved the most balanced results in terms of both detection accuracy and error area (see Table S3 in the Supplementary Information). These findings indicate a high level of practical applicability for object detection in operational contexts. Recent studies have reported that object detection models with an accuracy exceeding 90 % are considered suitable for real-world implementation [20,40,42].

The F1-Score exceeding 97 % achieved in this study significantly surpasses that benchmark. Furthermore, minimizing the error area is critical because an increased proportion of inaccurate detections leads to longer post-processing times. In the optimal configuration of this study, the error area ratio remained below 5 %, satisfying the practical threshold of > 95 % accurate detection area and < 5 % error area, which is considered essential for large-scale agricultural monitoring and mapping applications. The results demonstrate that the confidence score is the more influential parameter determining the overall level of detection accuracy, whereas the overlap ratio serves as a complementary parameter that controls boundary handling and detection stability under specific confidence conditions.

The final detection results of agricultural parcels obtained using the optimized YOLO–SAM configuration are presented in Fig. 9. Under these optimal parameters, the model achieved consistently high detection performance; however, detailed inspection of the segmentation outputs revealed minor geometric irregularities and noise in some areas. These artifacts are closely related to the prompting method used in the SAM model, particularly to the centroid-point prompting strategy adopted in this study. Therefore, further examination is required to discuss the underlying causes of these results and to validate the appropriateness of using centroid-point prompts for agricultural field segmentation.

3.4. Irregular noise of YOLO-SAM fusion model

After examining the paddy fields extracted through the YOLO-SAM fusion model, it was observed that some results displayed irregular noise, particularly around the centroid locations used by YOLO as input for segmentation (Fig. 10). While the SAM model automatically performs segmentation based on the input points, the accuracy varies depending on the prompt type and number. Although bbox prompts generally yield more stable segmentation for large objects, this study employed centroid-point prompts to better represent the geometry of irregular and fragmented paddy fields. In agricultural environments, bbox prompts can easily encompass non-agricultural areas such as paths, dikes, or adjacent fields, thereby introducing segmentation error. Using the centroid of the YOLO-detected bbox as a prompt allows the SAM model to focus on the core cultivated region while minimizing the influence of surrounding non-farming areas.

The accuracy can be improved by utilizing both positive and negative points. In this study, only the centroids of the predicted YOLO bboxes were used as positive points, which resulted in noise affecting the accuracy of the segmentation boundaries in some images. More positive points could be input to address this issue or negative points could be added to unnecessary locations. However, when using YOLO-SAM fusion model for irregularly shaped objects such as paddy fields, randomly adding positive and negative points could potentially input points at locations used for purposes other than agriculture. A drawback is that adding negative points would require manual input by the user.

To minimize the need for direct user input, this study implemented a semi-automated QA procedure in which the outputs extracted by YOLO-SAM fusion model were visually inspected and either accepted or rejected. On average, manual inspection required approximately 8–10 min per 100 fields, compared to 3–4 h typically needed for full manual polygon digitization. This process yielded 108 noise-free segmentation outputs, which were further augmented through vertical and horizontal flips and rotations of 90 and 15 degrees, resulting in 756 augmented segmentation training datasets. This approach ensures that the training

Fig. 9. YOLO-SAM Fusion Model with Optimized Parameters Configuration (Confidence score = 0.8, Overlap = 60 %). Green polygons represent the paddy field boundaries extracted by the YOLO–SAM fusion model, while red lines indicate the ground-truth reference boundaries. The model achieved an accuracy of 99.85 % of the actual cultivated area, with Precision = 97.13 %, Recall = 96.98 %, and F1-Score = 97.05 %.

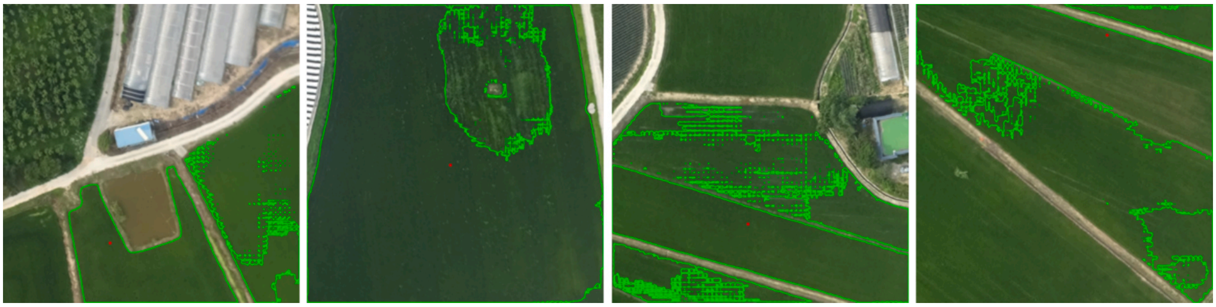

Fig. 10. Example of noisy data with segmentation information extracted by YOLO-SAM model. The figure shows representative cases where irregular noise and fragmented boundaries were observed around the centroid locations used as prompts in the YOLO–SAM segmentation process. These artifacts occurred primarily due to the use of centroid-point prompts, which sometimes generate excessive local segmentation in homogeneous areas.

data is both high quality and robust against variations in image orientation and layout. These optimized results provide a quantitative basis for comparison with existing GIS-based segmentation approaches, which is further discussed in the following section.

4. Discussion

This study proposed an automated object segmentation framework that integrates YOLO with SAM for accurate object boundary extraction. With the recent advancement of AI-based segmentation, several tools have been implemented in GIS platforms, among which the Deepness plugin for QGIS is one of the most widely used [1,26]. The comparison focuses on algorithmic performance under a consistent GIS environment, using the existing Deepness plugin as a reference implementation.

Because Deepness represents a widely adopted, operational YOLO-based segmentation approach within GIS environments, it provides a practical and relevant benchmark for evaluating the advancement and innovativeness of the proposed YOLO-SAM fusion framework. Deepness provides the advantage of direct integration with GIS data and supports multiple YOLO versions for object detection. However, for segmentation tasks, it requires polygon-based annotations and does not include a mechanism to automatically convert bounding boxes into segmentation masks, resulting in high data preparation costs. In contrast, the proposed YOLO–SAM fusion model achieves precise segmentation results using only bounding-box-level annotations. The proposed YOLO–SAM framework improves segmentation accuracy without altering the user-facing GIS workflow.

By exploiting the zero-shot capability of SAM, the model can be easily adapted to detect various objects such as rice paddies, buildings, roads, and forests by retraining only the YOLO detector, offering significantly higher versatility and practicality than object-specific segmentation models. For comparison between YOLO-SAM and QGIS Deepness plugin, segmentation data generated by YOLO–SAM were used to train a YOLO-Seg model, and the resulting weight files were applied within the Deepness plugin in QGIS. The YOLO–Seg model was trained using approximately 756 datasets constructed through the YOLO–SAM fusion model, divided into an 80:20 training-to-validation ratio, and trained with the YOLOv8m-Seg architecture for 200 epochs on a GPU. The model demonstrated stable convergence, reaching a precision of 0.858 and a recall of 0.628 (see Fig. S1 and context in Part S1 addressing YOLO-Seg training performance in the SI).

Both models used identical parameters with confidence = 0.5, IoU threshold = 0.5, overlap = 10 %, image resolution = 8 cm/px, and tile size = 1024 px. The YOLO–SAM detector was trained for 20 min and 20 s, whereas the YOLO-Seg training within Deepness required 41 min and 35 s, indicating that YOLO–SAM achieved approximately twice the training efficiency due to its reliance on bounding-box annotations rather than polygon masks. The operational GIS environment used for comparison with the Deepness plugin is illustrated in Fig. S1 of the SI. The convergence behavior observed in Fig. S1 supports this difference, showing that YOLO–SAM-derived datasets enabled faster stabilization of loss and higher mAP scores.

Quantitatively, YOLO–SAM achieved Precision = 97.13 %, Recall = 96.98 %, and F1-Score = 97.05 %, while Deepness achieved Precision = 95.55 %, Recall = 76.49 %, and F1-Score = 84.96 %. The recall difference of 20.49 percentage points indicates that YOLO–SAM was substantially more effective at capturing true agricultural boundaries. As summarized in Table S3, the YOLO–SAM model achieved a mean F1-Score exceeding 97 % under the optimized parameters (Confidence = 0.8, Overlap = 60 %), demonstrating high boundary accuracy and consistency. When these results are compared with the Deepness plugin output, the improvement in recall (from 76.49 % to 96.98 %) and reduction of false negatives (Fig. 8) clearly indicate the superior detection capability of the proposed method.

In error-area analysis, YOLO–SAM recorded FN = 7,070.69 m^2 (3.0 %) and FP = 6,708.40 m^2, whereas Deepness yielded FN = 54,936.19 m^2 (23.5 %) and FP = 8,329.97 m^2, confirming that YOLO–SAM reduced undetected areas by a factor of approximately 7.8 times. This finding is consistent with the boundary-level accuracy illustrated in Fig. 9, where YOLO–SAM more precisely delineates paddy field edges with minimal omission. Although Deepness processed inference faster (6 min 50 s) than YOLO–SAM (13 min 20 s) due to its single-stage structure, the proposed model demonstrated superior boundary accuracy and overall reliability (see Table S4 in the SI for a quantitative performance comparison of the YOLO-SAM model and Deepness (YOLO-Seg)).

A qualitative comparison also revealed distinct differences in boundary representation. As further visualized in Fig. 10, the centroid-based prompt used in YOLO–SAM preserves curved and fragmented boundaries, while Deepness tends to simplify them into straight lines. In Deepness, agricultural fields with complex or curved geometries were often simplified into straight-edged polygons (Fig. S1), likely due to smoothing or simplification algorithms designed to reduce data size. This caused boundary distortion, leading to potential area overestimation or underestimation. Such limitations can be critical in applications requiring high spatial precision, such as cadastral boundary definition, agricultural compensation, and precision-farming parcel management. In contrast, the YOLO–SAM fusion model, leveraging SAM's high-resolution segmentation capability, accurately captured fine curvatures and boundary details, thereby enhancing both quantitative accuracy and practical applicability.

Overall, the YOLO–SAM framework consistently outperformed Deepness across quantitative (Table S4) and qualitative (Figs. 9–10) assessments, confirming its robustness for practical GIS-based applications. Despite its advantages, the YOLO–SAM model has several limitations. First, due to its two-stage structure (YOLO detection followed by SAM segmentation), the analysis time is approximately twice that of Deepness, limiting real-time applicability. Second, the current model was validated only on paddy fields from a single region, and further evaluation is needed for diverse crop types, climatic conditions, and seasonal variations. Third, SAM was used as a pre-trained general model without domain-specific fine-tuning for agricultural imagery.

Future work will address these issues through model optimization and domain adaptation: (1) improving processing speed via model compression and GPU optimization, (2) expanding validation to diverse agricultural environments, (3) fine-tuning SAM for agricultural imagery to enhance domain-specific accuracy, (4) extending the approach to multi-temporal satellite imagery for crop monitoring and land-use change detection, and (5) developing full integration with GIS platforms such as QGIS for large-scale operational deployment.

5. Conclusion

We developed a YOLO-SAM fusion framework that integrates YOLO's semantic object detection with SAM's geometric segmentation to automate agricultural land delineation from high-resolution drone imagery. The proposed approach achieved high segmentation accuracy (F1 = 97.05 %), while reducing manual labeling time from approximately 3–4 h for full manual polygon digitization to 8–10 min per 100 fields using the YOLO–SAM fusion model with semi-automated QA, demonstrating strong potential for efficient agricultural monitoring.

The framework also proved scalable and adaptable, producing precise parcel boundaries that can be directly utilized in GIS environments such as QGIS. Although the model achieved excellent accuracy, its validation was limited to a single agricultural district, suggesting that additional testing across multiple regions and crop types is required to confirm generalizability. Future work will refine this vectorization process, such as by adaptive tolerance control, to better preserve irregular field contours while maintaining manageable file sizes.

Overall, the YOLO–SAM fusion framework provides a cost-effective, accurate, and operationally practical solution for automated agricultural land detection, advancing both methodological and applied aspects of precision agriculture and geospatial analysis. This study contributes a valuable tool to agricultural monitoring, particularly through its potential integration with platforms such as QGIS.

CRediT authorship contribution statement

Jeongbae Jeon: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Solhee Kim:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Taegon Kim:** Writing – review & editing, Validation, Supervision, Software, Project administration, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.inpa.2025.12.008.

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